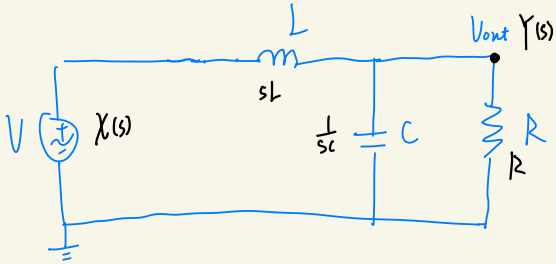


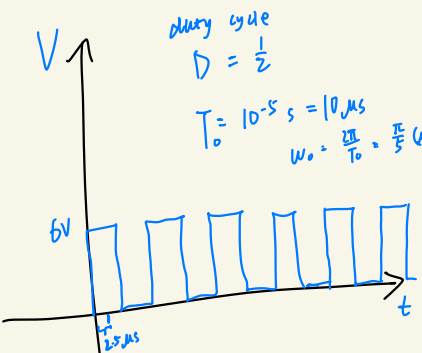
没有二极管求解：

$V(s)$ ✓

$H(s)$ ✓

$\Rightarrow Y(s) = X(s) \cdot H(s)$

Laplace：



duty cycle

$D = \frac{1}{2}$

$T_0 = 10^{-5}\,s = 10\,\mu s$

$\omega_0 = \frac{2\pi}{T_0} = \frac{\pi}{5}\,(\mu s)^{-1}$

$$X(s) = \mathcal{L}[V(t)] = \frac{6}{s} \cdot \frac{e^{-5s}-1}{e^{-10s}-1}$$

$$H(s) = \frac{\dfrac{\frac{R}{sC}}{\frac{1}{sC}+R}}{\dfrac{\frac{R}{sC}}{\frac{1}{sC}+R} + sL} = \frac{R}{R+sL+s^2CLR}$$

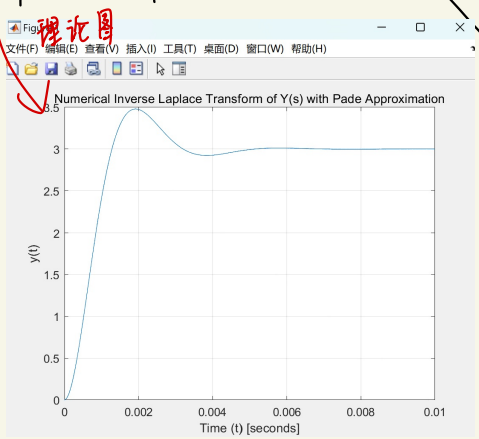
$$H(s) = \frac{150}{150+0.08s+1.2\times10^{-5}s^2}$$

$$\therefore Y(s) = \frac{6}{s} \cdot \frac{e^{-5\times10^{-6}s}-1}{e^{-10\times10^{-6}s}-1} \cdot \frac{150}{150+0.08s+1.2\times10^{-5}s^2}$$

if $L = 80mH$, $C = 3.5\,\mu F$, $R = 150\,\Omega$

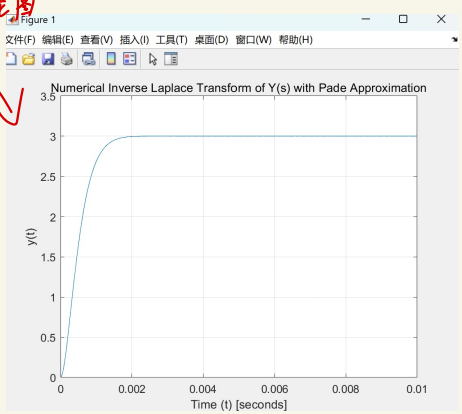
$$Y(s) = \cdots \frac{150}{150+0.08s+4.2\times10^{-5}s^2}$$

理论图



if $L = 80mH$, $C = 1\,\mu F$, $R = 150\,\Omega$

理论图



符合 仿真结果